Chapter 2. Describing Contingency Tables

Proportions, relative risk, odds ratios, partial association

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2.1 Probability structure for contingency tables

Motivation. The joint distribution contains the whole relationship; the sampling design decides which distributions a table can estimate.

2.1.1 Contingency tables and their distributions

Let X have I and Y have J categories. An $I \times J$ table of cell counts is a *contingency table* (Pearson, 1904).

Definition (Joint, marginal and conditional distributions)

(a) The *joint distribution* is $\{\pi_{ij}\}$, $\pi_{ij} = P(X = i, Y = j)$.
(b) The *marginal distributions* are $\pi_{i+} = \sum_j \pi_{ij}$ and $\pi_{+j} = \sum_i \pi_{ij}$, with $\sum_i \pi_{i+} = \sum_j \pi_{+j} = \sum_i \sum_j \pi_{ij} = 1$ (a "+" denotes summation over that index).
(c) When Y is a response, the *conditional distribution* of Y in row i is $\pi_{j|i} = P(Y = j \mid X = i) = \pi_{ij}/\pi_{i+}$, with $\sum_j \pi_{j|i} = 1$.

If X is fixed by design, only the conditional distributions are meaningful; studies typically compare those of Y across levels of X.

Example 2.1 (Aspirin and heart attacks)

The Physicians' Health Study: a five-year randomized blind trial of aspirin versus placebo (every other day) for cardiovascular mortality.

	Myocardial infarction			
Group	Fatal	Nonfatal	No attack	Total
Placebo	18	171	10,845	11,034
Aspirin	5	99	10,933	11,037

Of 11,034 placebo takers 18 had fatal attacks, against 5 of 11,037 aspirin takers.

2.1.2 Sensitivity and specificity

For a diagnostic test let X = true disease status and Y = diagnosis (positive/negative).

Definition (Sensitivity, specificity)

Sensitivity $= P(Y = \text{positive} \mid X = \text{diseased}) = \pi_{1|1}$; *specificity* $= P(Y = \text{negative} \mid X = \text{not diseased}) = \pi_{2|2}$.

Mammography with clinical examination: sensitivity 0.82, specificity 0.99 (literature survey).

2.1.3 Independence of categorical variables

Definition (Independence)

X and Y are *independent* if $\pi_{ij} = \pi_{i+}\pi_{+j}$ for all i, j.

Proposition 2.2 (Independence as homogeneity)

If all $\pi_{i+} > 0$, then X and Y are independent if and only if $\pi_{j|1} = \pi_{j|2} = \cdots = \pi_{j|I}$ for every j; the common value is then π_{+j}.

Proof.

Step 1. If $\pi_{ij} = \pi_{i+}\pi_{+j}$ then $\pi_{j|i} = \pi_{ij}/\pi_{i+} = \pi_{+j}$ for every i.

Step 2. Conversely, if $\pi_{j|i} = c_j$ for all i, then $\pi_{+j} = \sum_i \pi_{i+}\pi_{j|i} = c_j \sum_i \pi_{i+} = c_j$, so $\pi_{ij} = \pi_{i+}\pi_{j|i} = \pi_{i+}\pi_{+j}$. ■

For a response Y this *homogeneity of conditional distributions* is the natural definition. Sample analogues: $p_{ij} = n_{ij}/n$, $p_{j|i} = n_{ij}/n_{i+}$, $n = \sum_i \sum_j n_{ij}$ (also written $\hat{\pi}_{ij}$).

2.1.4 Poisson, binomial, and multinomial sampling

(a) *Poisson sampling*: the IJ counts Y_{ij} are independent Poisson(μ_{ij}), with joint pmf $\prod_i \prod_j e^{-\mu_{ij}} \mu_{ij}^{n_{ij}}/n_{ij}!$; nothing is fixed.

(b) *Multinomial sampling*: n is fixed and $\{n_{ij}\}$ is multinomial over the IJ cells, $\frac{n!}{\prod\prod n_{ij}!} \prod\prod \pi_{ij}^{n_{ij}}$.

(c) *Independent (product) multinomial sampling*: the row totals $n_i = n_{i+}$ are fixed and each row is an independent multinomial sample with probabilities $\{\pi_{j|i}\}$:

$$\prod_i \Big[\frac{n_i!}{\prod_j n_{ij}!} \prod_j \pi_{j|i}^{n_{ij}} \Big]. \tag{2.1}$$

(d) When both margins are fixed the distribution is hypergeometric (Section 3.5).

Theorem 2.3 (Conditioning on row totals gives product multinomial)

If $\{n_{ij}\}$ arise from independent Poisson sampling with means $\{\mu_{ij}\}$, or from multinomial sampling with probabilities $\pi_{ij} = \mu_{ij}/n$, then conditional on the row totals $\{n_{i+}\}$ the rows are independent multinomial samples with probabilities $\pi_{j|i} = \mu_{ij}/\mu_{i+}$.

Proof. [Supplement]

Step 1 (Poisson). By Theorem 1.5 applied to row i, given n_{i+} the counts $(n_{i1}, \ldots, n_{iJ})$ are multinomial($n_{i+}, \{\mu_{ij}/\mu_{i+}\}$). Each row's counts and total involve only that row's Poisson variables, so the rows stay independent given all row totals.

Step 2 (multinomial). The row totals $(n_{1+}, \ldots, n_{I+})$ are multinomial($n, \{\pi_{i+}\}$) (collapsing categories of a multinomial). Hence for tables with these row totals,

$$P(\{n_{ij}\} \mid \{n_{i+}\}) = \frac{\frac{n!}{\prod_i \prod_j n_{ij}!} \prod_i \prod_j \pi_{ij}^{n_{ij}}}{\frac{n!}{\prod_i n_{i+}!} \prod_i \pi_{i+}^{n_{i+}}} = \prod_i \Big[\frac{n_{i+}!}{\prod_j n_{ij}!} \prod_j \Big(\frac{\pi_{ij}}{\pi_{i+}} \Big)^{n_{ij}} \Big],$$

using $\pi_{i+}^{n_{i+}} = \prod_j \pi_{i+}^{n_{ij}}$. This is (2.1) with $\pi_{j|i} = \pi_{ij}/\pi_{i+} = \mu_{ij}/\mu_{i+}$. ■

So for explanatory X one may analyse conditional on the row totals even when the design did not fix them.

2.1.5 Seat belt example

Example 2.4 (Seat belts and crash outcome)

Seat-belt use (yes/no) by crash outcome (fatal/nonfatal), Massachusetts Turnpike:
(a) recording all accidents in the coming year: n random, the four counts independent Poisson;
(b) sampling 200 police records from last year: $n = 200$ fixed, multinomial over four cells;
(c) sampling 100 fatal and 100 nonfatal records: column totals fixed, each column an independent binomial;
(d) the (unethical) experiment randomizing 100 of 200 subjects to wear belts and forcing crashes: row totals fixed, independent binomial rows.

2.1.6 Types of studies

Example 2.5 (Smoking and lung cancer: a case–control study)

Doll and Hill: 709 lung cancer patients in 20 London hospitals, each with a matched noncancer control (hospital, gender, 5-year age group), asked about smoking.

	Lung cancer	
Smoker	Cases	Controls
Yes	688	650
No	21	59
Total	709	709

The cancer margin is fixed: $P(\text{smoker} \mid \text{case}) = 688/709 = 0.970$, $P(\text{smoker} \mid \text{control}) = 650/709 = 0.917$; $P(\text{cancer} \mid \text{smoking})$ needs the prevalence (Bayes).

Definition (Study designs)

(a) *Retrospective* (case–control): samples at fixed levels of Y, X looked up in the past; each column is multinomial on X.
(b) *Prospective*: groups defined by X are followed up (*clinical trials* randomize X, *cohort studies* do not); each row is multinomial on Y.
(c) *Cross-sectional*: subjects classified on both variables at once; one multinomial sample of IJ counts.

Case–control, cohort and cross-sectional studies are *observational* (more prone to bias); a clinical trial is *experimental*, randomization balancing other variables.

2.2 Comparing two proportions

Motivation. Difference, ratio and odds ratio compare π_1, π_2 and behave very differently under change of sampling design.

In a 2×2 table write $\pi_i = \pi_{1|i}$ for the success probability in row i, so $\pi_{2|i} = 1 - \pi_i$.

2.2.1 Difference of proportions

The *difference of proportions* $\pi_1 - \pi_2 \in [-1, 1]$ is the same for successes or failures up to sign, and is 0 iff Y is independent of the row classification. The column difference is in general different.

2.2.2 Relative risk

A difference of fixed size may matter more near 0 or 1: $0.010 - 0.001$ and $0.410 - 0.401$ are both 0.009, but the ratios are 10.0 and 1.02.

Definition (Relative risk)

The *relative risk* is $\pi_1/\pi_2 \in [0, \infty)$; it equals 1 under independence. Comparing the second response category gives a different value $(1 - \pi_1)/(1 - \pi_2)$.

2.2.3 Odds ratio

Definition (Odds, odds ratio)

For success probability π the *odds* are $\Omega = \pi/(1 - \pi)$, inverted by $\pi = \Omega/(\Omega + 1)$. With $\Omega_i = \pi_i/(1 - \pi_i) = \pi_{i1}/\pi_{i2}$, the *odds ratio* is

$$\theta = \frac{\Omega_1}{\Omega_2} = \frac{\pi_1/(1 - \pi_1)}{\pi_2/(1 - \pi_2)} = \frac{\pi_{11}\pi_{22}}{\pi_{12}\pi_{21}}, \tag{2.2}$$

also called the *cross-product ratio* (Yule). The sample odds ratio is $\hat{\theta} = n_{11}n_{22}/(n_{12}n_{21})$.

E.g. $\pi = 0.75$ gives $\Omega = 3$: a success is three times as likely as a failure; $\Omega = 1/3$ gives $\pi = 0.25$.

2.2.4 Properties of the odds ratio

Proposition 2.6 (Properties of the odds ratio)

Assume all $\pi_{ij} > 0$.
(a) $\theta = 1$ iff $\Omega_1 = \Omega_2$ iff $\pi_1 = \pi_2$ (independence); $\theta > 1$ iff $\pi_1 > \pi_2$.
(b) Reversing the order of the rows (or of the columns) replaces θ by $1/\theta$, i.e. $\log\theta$ by $-\log\theta$; transposing the table leaves θ unchanged.
(c) θ is the same whichever conditional direction is used:

$$\theta = \frac{P(Y = 1 \mid X = 1)/P(Y = 2 \mid X = 1)}{P(Y = 1 \mid X = 2)/P(Y = 2 \mid X = 2)} = \frac{P(X = 1 \mid Y = 1)/P(X = 2 \mid Y = 1)}{P(X = 1 \mid Y = 2)/P(X = 2 \mid Y = 2)}. \tag{2.3}$$

(d) $\hat{\theta}$ is unchanged when both counts in a row, or both counts in a column, are multiplied by a positive constant.

Proof.

Step (a). $x \mapsto x/(1 - x)$ is strictly increasing on $(0, 1)$, so $\Omega_1 \gtreqless \Omega_2$ iff $\pi_1 \gtreqless \pi_2$.

Step (b). Swapping rows maps $\pi_{11}\pi_{22}/(\pi_{12}\pi_{21})$ to $\pi_{21}\pi_{12}/(\pi_{22}\pi_{11}) = 1/\theta$; similarly for columns. Transposing swaps π_{12} and π_{21}, which leaves the cross-product ratio unchanged.

Step (c). The first ratio is $(\pi_{11}/\pi_{12})/(\pi_{21}/\pi_{22})$ after cancelling π_{1+} and π_{2+}; the second is $(\pi_{11}/\pi_{21})/(\pi_{12}/\pi_{22})$ after cancelling π_{+1}, π_{+2}. Both equal $\pi_{11}\pi_{22}/(\pi_{12}\pi_{21})$.

Step (d). Multiplying row 1 by c multiplies numerator $n_{11}n_{22}$ and denominator $n_{12}n_{21}$ each by c; columns likewise. ■

$\theta = 4$ means the odds (not the probability) in row 1 are four times those in row 2; $\log 4 = 1.39$ and $\log 0.25 = -1.39$ are equally strong. By (c)–(d), $\hat{\theta}$ estimates the same parameter in prospective, retrospective and cross-sectional designs (e.g. $100 + 100$ or $40 + 160$ in a case–control study). The sample difference and relative risk are invariant to scaling rows only, not columns or transposition.

2.2.5 Aspirin and heart attacks revisited

Example 2.7 (Three measures for the aspirin study)

Combining fatal and nonfatal attacks in Example 2.1: placebo $189/11{,}034 = 0.0171$, aspirin $104/11{,}037 = 0.0094$.
(a) Difference: $0.0171 - 0.0094 = 0.0077$.
(b) Relative risk: $0.0171/0.0094 = 1.82$.
(c) Odds ratio: $\hat{\theta} = \dfrac{189 \times 10{,}933}{10{,}845 \times 104} = 1.83$; the odds of heart attack were 83% higher with placebo.

2.2.6 Case–control studies and the odds ratio

A case–control study estimates $P(X = i \mid Y = j)$, not $P(Y = j \mid X = i)$, hence neither the difference nor the relative risk; by (2.3) the odds ratio is the same in either direction.

Example 2.8 (Odds ratio from Doll and Hill's data)

From Example 2.5,

$$\hat{\theta} = \frac{(688/709)/(21/709)}{(650/709)/(59/709)} = \frac{688 \times 59}{650 \times 21} = 3.0.$$

Estimated odds of lung cancer for smokers: 3.0 times those for nonsmokers.

2.2.7 Relationship between odds ratio and relative risk

Proposition 2.9 (Odds ratio versus relative risk)

$$\theta = \frac{\pi_1}{\pi_2} \times \frac{1 - \pi_2}{1 - \pi_1}.$$

So $\theta \approx$ relative risk when π_1, π_2 are both near 0, and θ is farther from 1 (same side).

Proof.

Step 1. Directly from (2.2): $\theta = \frac{\pi_1}{1-\pi_1} \cdot \frac{1-\pi_2}{\pi_2}$.

Step 2. If $\pi_1 > \pi_2$ then $(1 - \pi_2)/(1 - \pi_1) > 1$ so $\theta > \pi_1/\pi_2 > 1$; if $\pi_1 < \pi_2$ the factor is < 1 and $\theta < \pi_1/\pi_2 < 1$. As $\pi_1, \pi_2 \to 0$ the factor tends to 1. ■

Aspirin study ($\pi_i < 0.02$): relative risk 1.82, odds ratio 1.83. So for rare diseases $\hat{\theta}$ estimates the relative risk (Cornfield, 1951): smokers had about 3 times the lung cancer rate.

2.3 Partial association in stratified 2×2 tables

Motivation. A third variable Z affecting both X and Y can change, even reverse, an apparent association once held fixed.

E.g. a passive-smoking study must control age and socioeconomic status: spouses of nonsmokers may be younger, and the young get lung cancer less often.

2.3.1 Partial tables

Definition (Partial and marginal tables)

For fixed Z, the two-way cross-sections of an $X \times Y \times Z$ table are *partial tables* with *conditional associations*; summing them gives the XY *marginal table*, which ignores Z.

2.3.2 Death penalty example

Example 2.10 (Death penalty verdict by defendant's and victims' race)

674 defendants in Florida multiple-murder indictments, 1976–1987. Y = death penalty, X = defendant's race, Z = victims' race.

Victims' race	Defendant's race	Yes	No	Percent yes
White	White	53	414	11.3
	Black	11	37	22.9
Black	White	0	16	0.0
	Black	4	139	2.8
Total	White	53	430	11.0
	Black	15	176	7.9

(a) Controlling for victims' race, black defendants received the death penalty more often: by $22.9 - 11.3 = 11.6$ points with white victims and 2.8 points with black victims.
(b) Ignoring victims' race, white defendants received it more often (11.0% vs. 7.9%): the association reverses.
(c) Victims' race is strongly associated with defendant's race (marginal odds ratio $467 \times 143/(48 \times 16) = 87.0$), and the death penalty is much more likely for white victims.

Definition (Simpson's paradox)

A marginal association whose direction differs from that of each conditional association is an instance of *Simpson's paradox* (Yule 1903, Simpson 1951).

It warns against causal readings of an X–Y association. Against Fisher's genetic-factor objection, Cornfield showed that a confounder explaining a very strong association must itself be very strongly associated with X and Y.

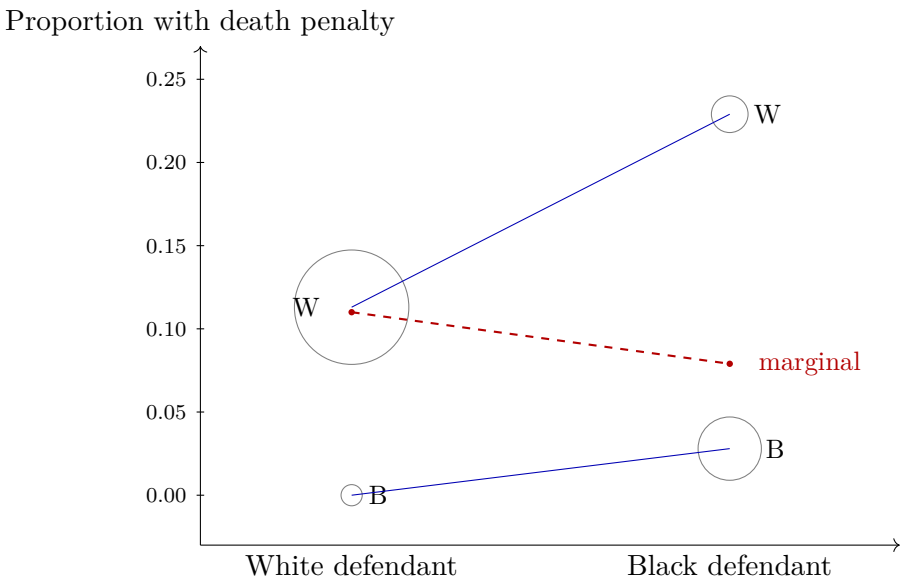


Figure 1: Simpson's paradox in the death penalty data (letters: victims' race; area $\propto$ cases). Conditional lines rise, the marginal line falls.

2.3.3 Conditional and marginal odds ratios

For a $2 \times 2 \times K$ table with expected frequencies μ_{ijk}:

$$\theta_{XY(k)} = \frac{\mu_{11k}\mu_{22k}}{\mu_{12k}\mu_{21k}} \quad \text{(conditional)}, \qquad \theta_{XY} = \frac{\mu_{11+}\mu_{22+}}{\mu_{12+}\mu_{21+}} \quad \text{(marginal)},$$

with $\mu_{ij+} = \sum_k \mu_{ijk}$; sample versions use counts.

Example 2.11 (Odds ratios for the death penalty data)

(a) White victims: $\hat{\theta}_{XY(1)} = \dfrac{53 \times 37}{414 \times 11} = 0.43$; the odds of the death penalty for white defendants were 43% of those for black defendants.
(b) Black victims: $\hat{\theta}_{XY(2)} = \dfrac{0 \times 139}{16 \times 4} = 0.0$ (no white defendant with black victims got the death penalty).
(c) Marginal: $\hat{\theta}_{XY} = \dfrac{53 \times 176}{430 \times 15} = 1.45$; the odds were 45% higher for white defendants.

2.3.4 Marginal versus conditional independence

Definition (Conditional independence)

X and Y are *conditionally independent at level* k of Z if they are independent in partial table k, i.e. $P(Y = j \mid X = i, Z = k) = P(Y = j \mid Z = k)$ for all i, j. They are *conditionally independent given* Z if this holds for every k.

Proposition 2.12 (Factorization under conditional independence)

For a single multinomial over the three-way table with $\pi_{ijk} = P(X = i, Y = j, Z = k)$ and all $\pi_{++k} > 0$, X and Y are conditionally independent given Z iff

$$\pi_{ijk} = \frac{\pi_{i+k}\,\pi_{+jk}}{\pi_{++k}} \qquad \text{for all } i, j, k.$$

Conditional independence does not imply marginal independence: in general $\pi_{ij+} = \sum_k \pi_{i+k}\pi_{+jk}/\pi_{++k} \neq \pi_{i++}\pi_{+j+}$.

Proof.

Step 1. Always $\pi_{ijk} = P(X = i, Z = k)\,P(Y = j \mid X = i, Z = k) = \pi_{i+k}\,P(Y = j \mid X = i, Z = k)$.

Step 2. Under conditional independence the last factor equals $P(Y = j \mid Z = k) = \pi_{+jk}/\pi_{++k}$, giving the factorization. Conversely, the factorization gives $P(Y = j \mid X = i, Z = k) = \pi_{ijk}/\pi_{i+k} = \pi_{+jk}/\pi_{++k}$ whenever $\pi_{i+k} > 0$.

Step 3. Summing over k gives $\pi_{ij+} = \sum_k \pi_{i+k}\pi_{+jk}/\pi_{++k}$, which involves k in every factor and does not reduce to $\pi_{i++}\pi_{+j+}$; Example 2.13 is an explicit counterexample. ■

For $2 \times 2 \times K$ tables, conditional independence means $\theta_{XY(k)} = 1$ for all k.

Example 2.13 (Conditional but not marginal independence)

Expected frequencies for Y = response, X = treatment, Z = clinic:

Clinic	Treatment	Success	Failure
1	A	18	12
	B	12	8
2	A	2	8
	B	8	32
Total	A	20	20
	B	20	40

(a) $\theta_{XY(1)} = \dfrac{18 \times 8}{12 \times 12} = 1.0$ and $\theta_{XY(2)} = \dfrac{2 \times 32}{8 \times 8} = 1.0$: conditional independence.

(b) Marginal: $\theta_{XY} = \dfrac{20 \times 40}{20 \times 20} = 2.0$.

(c) All XZ and YZ conditional odds ratios are 6.0, e.g. $\dfrac{18 \times 8}{12 \times 2}$: clinic 1 uses A more and has more successes, so the marginal table misleads.

2.3.5 Homogeneous association

Definition (Homogeneous association)

A $2 \times 2 \times K$ table has *homogeneous XY association* if $\theta_{XY(1)} = \cdots = \theta_{XY(K)}$; conditional independence is the case of common value 1. Otherwise there is *interaction* and Z is an *effect modifier*.

Proposition 2.14 (Homogeneous association is symmetric)

For a $2 \times 2 \times 2$ table with positive probabilities,

$$\frac{\theta_{XY(1)}}{\theta_{XY(2)}} = \frac{\theta_{XZ(1)}}{\theta_{XZ(2)}} = \frac{\theta_{YZ(1)}}{\theta_{YZ(2)}},$$

where $\theta_{XZ(j)}$ is the XZ odds ratio at level j of Y and $\theta_{YZ(i)}$ the YZ odds ratio at level i of X. So, for $2 \times 2 \times K$, homogeneous XY association holds iff for all k, k' the XZ (and YZ) odds ratio formed with k, k' is the same at both levels of Y (of X).

Proof. [Supplement]

Step 1. Directly,

$$\frac{\theta_{XY(1)}}{\theta_{XY(2)}} = \frac{\pi_{111}\pi_{221}}{\pi_{121}\pi_{211}} \cdot \frac{\pi_{122}\pi_{212}}{\pi_{112}\pi_{222}}, \qquad \frac{\theta_{XZ(1)}}{\theta_{XZ(2)}} = \frac{\pi_{111}\pi_{212}}{\pi_{112}\pi_{211}} \cdot \frac{\pi_{122}\pi_{221}}{\pi_{121}\pi_{222}}.$$

Both equal $\dfrac{\pi_{111}\pi_{122}\pi_{212}\pi_{221}}{\pi_{112}\pi_{121}\pi_{211}\pi_{222}}$ (odd index sums over even), symmetric in the three indices, so it also equals $\theta_{YZ(1)}/\theta_{YZ(2)}$.

Step 2. For $2 \times 2 \times K$, apply Step 1 to the $2 \times 2 \times 2$ subtable formed by levels k, k' of Z: $\theta_{XY(k)} = \theta_{XY(k')}$ iff the XZ odds ratio for (k, k') is the same at $Y = 1$ and $Y = 2$, and likewise for YZ. ■

Homogeneity is thus a joint property ("no three-factor interaction"). Odds ratios 1.2, 3.9, 8.8 in age groups < 45, 45–65, > 65 would make age an effect modifier. Death penalty: $\hat{\theta}_{XY(1)} = 0.43$, $\hat{\theta}_{XY(2)} = 0.0$ (zero count; 0.94 after adding 1/2), not contradicting homogeneity (tests in Section 6.3).

2.4 Extensions for $I \times J$ tables

Motivation. Beyond 2×2, a minimal set of odds ratios describes association completely; summary indices trade completeness for interpretability.

2.4.1 Odds ratios in $I \times J$ tables

Any pair of rows a, b and columns c, d gives an odds ratio $\pi_{ac}\pi_{bd}/(\pi_{bc}\pi_{ad})$; there are $\binom{I}{2}\binom{J}{2}$ of them, highly redundant.

Definition (Local and baseline odds ratios)

The *local odds ratios* are

$$\theta_{ij} = \frac{\pi_{ij}\,\pi_{i+1,j+1}}{\pi_{i,j+1}\,\pi_{i+1,j}}, \qquad i = 1, \ldots, I-1,\ j = 1, \ldots, J-1. \tag{2.4}$$

An alternative basic set uses the last row and column as baseline:

$$\alpha_{ij} = \frac{\pi_{ij}\,\pi_{IJ}}{\pi_{Ij}\,\pi_{iJ}}, \qquad i = 1, \ldots, I-1,\ j = 1, \ldots, J-1. \tag{2.5}$$

Proposition 2.15 ($(I-1)(J-1)$ odds ratios describe the association)

Assume all $\pi_{ij} > 0$.
(a) Every odds ratio $\pi_{ac}\pi_{bd}/(\pi_{bc}\pi_{ad})$ with $a < b$, $c < d$ is the product $\prod_{i=a}^{b-1}\prod_{j=c}^{d-1}\theta_{ij}$.
(b) Given the marginal distributions $\{\pi_{i+}\}, \{\pi_{+j}\}$, the local odds ratios determine the cell probabilities uniquely.
(c) Independence holds iff all $(I-1)(J-1)$ local odds ratios equal 1.

Proof. [Supplement]

Step (a). Let $\lambda_{ij} = \log \pi_{ij}$ and $\log \theta_{ij} = \lambda_{ij} - \lambda_{i,j+1} - \lambda_{i+1,j} + \lambda_{i+1,j+1}$. Summing over $j = c, \ldots, d-1$ telescopes to $\lambda_{ic} - \lambda_{id} - \lambda_{i+1,c} + \lambda_{i+1,d}$; summing that over $i = a, \ldots, b-1$ telescopes to $\lambda_{ac} - \lambda_{ad} - \lambda_{bc} + \lambda_{bd}$, the log of the stated odds ratio.

Step (b). Suppose $\{\pi_{ij}\}$ and $\{\pi'_{ij}\}$ have the same margins and the same local odds ratios. Let $\delta_{ij} = \log(\pi'_{ij}/\pi_{ij})$. Its local contrasts $\delta_{ij} - \delta_{i,j+1} - \delta_{i+1,j} + \delta_{i+1,j+1}$ vanish, so by the telescoping in (a) so does $\delta_{ij} - \delta_{iJ} - \delta_{Ij} + \delta_{IJ}$. With $r_i = \delta_{iJ} - \delta_{IJ}$ and $c_j = \delta_{Ij}$ this gives $\delta_{ij} = r_i + c_j$. Hence

$$\sum_i \sum_j (\pi'_{ij} - \pi_{ij}) \log \frac{\pi'_{ij}}{\pi_{ij}} = \sum_i r_i(\pi'_{i+} - \pi_{i+}) + \sum_j c_j(\pi'_{+j} - \pi_{+j}) = 0.$$

Each term $(\pi' - \pi)\log(\pi'/\pi)$ is ≥ 0, with equality iff $\pi' = \pi$ (log is increasing). Hence $\pi'_{ij} = \pi_{ij}$ for all i, j.

Step (c). If $\pi_{ij} = \pi_{i+}\pi_{+j}$ then every $\theta_{ij} = 1$. Conversely, the independence table $\{\pi_{i+}\pi_{+j}\}$ has the same margins and all local odds ratios 1; by (b) it is the only such table. ■

So $(I-1)(J-1)$ parameters describe any $I \times J$ association. Example 2.1: local odds ratios $\frac{18\times 99}{171\times 5} = 2.08$ (fatal vs. nonfatal) and $\frac{171\times 10{,}933}{99\times 10{,}845} = 1.74$ (nonfatal vs. none), product 3.63 (first vs. third column); the more serious outcome was more prevalent under placebo. For $I \times J \times K$, homogeneous XY association means each conditional XY odds ratio is constant in Z.

2.4.2 Summary measures of association

A *proportional reduction in variation* measure, with $V(Y)$ a variation measure for the marginal distribution $\{\pi_{+j}\}$ and $V(Y \mid i)$ that for $\{\pi_{j|i}\}$, is

$$\frac{V(Y) - E[V(Y \mid X)]}{V(Y)}, \qquad E[V(Y \mid X)] = \sum_i \pi_{i+} V(Y \mid i), \tag{2.6}$$

generalizing R^2. For a nominal response Theil used the *entropy* $V(Y) = -\sum_j \pi_{+j} \log \pi_{+j}$, giving the *uncertainty coefficient*

$$U = -\frac{\sum_i \sum_j \pi_{ij} \log(\pi_{ij}/(\pi_{i+}\pi_{+j}))}{\sum_j \pi_{+j} \log \pi_{+j}}. \tag{2.7}$$

Proposition 2.16 (Range of the uncertainty coefficient)

If more than one $\pi_{+j} > 0$, then $0 \leq U \leq 1$; $U = 0$ iff X and Y are independent, and $U = 1$ iff for each i some $\pi_{j|i} = 1$.

Proof. [Supplement] Write

$$H(Y) = -\sum_j \pi_{+j} \log \pi_{+j} > 0, \qquad H(Y \mid X) = -\sum_i \pi_{i+} \sum_j \pi_{j|i} \log \pi_{j|i} \geq 0.$$

Step 1. (2.7) is (2.6) with entropy: since $\log \pi_{j|i} = \log \pi_{ij} - \log \pi_{i+}$, $H(Y) - H(Y \mid X) = \sum_i \sum_j \pi_{ij} \log \frac{\pi_{ij}}{\pi_{i+}\pi_{+j}}$, so $U = [H(Y) - H(Y \mid X)]/H(Y)$.

Step 2. By Jensen, $-\sum\sum \pi_{ij} \log \frac{\pi_{i+}\pi_{+j}}{\pi_{ij}} \geq -\log \sum\sum \pi_{i+}\pi_{+j} \geq 0$, with equality iff $\pi_{ij} = \pi_{i+}\pi_{+j}$ on all cells (then the sum is 1). So $U \geq 0$, with equality iff independence.

Step 3. $U \leq 1$ iff $H(Y \mid X) \geq 0$, true since each $-\pi \log \pi \geq 0$; equality iff every conditional distribution is degenerate. ■

Such measures are hard to interpret (a "30% reduction in entropy"?); ordinal summaries are easier.

2.4.3 Ordinal trends: concordant and discordant pairs

For ordinal X, Y, a pair is *concordant* if the subject higher on X is also higher on Y, *discordant* if lower, and *tied* if they agree on X and/or Y.

Example 2.17 (Job satisfaction and income)

Black males, 1996 General Social Survey ($n = 96$); VD/LD/MS/VS = very / a little dissatisfied, moderately / very satisfied.

Income ($)	VD	LD	MS	VS
< 15,000	1	3	10	6
15,000–25,000	2	3	10	7
25,000–40,000	1	6	14	12
> 40,000	0	1	9	11

Each cell's count times the total count below and to the right gives concordant pairs:

$$\begin{aligned} C = {} & 1(3+10+7+6+14+12+1+9+11) + 3(10+7+14+12+9+11) \\ & + 10(7+12+11) + 2(6+14+12+1+9+11) + 3(14+12+9+11) + 10(12+11) \\ & + 1(1+9+11) + 6(9+11) + 14(11) = 1331, \end{aligned}$$

and similarly (below and to the left) $D = 3(2+1+0)+10(2+3+1+6+0+1)+\cdots+12(0+1+9) = 849$. Since $C > D$, low income tends to occur with low satisfaction.

For two independent observations from $\{\pi_{ij}\}$ the probabilities of concordance and discordance are

$$\Pi_c = 2\sum_i \sum_j \pi_{ij}\Big(\sum_{h>i}\sum_{k>j} \pi_{hk}\Big), \qquad \Pi_d = 2\sum_i \sum_j \pi_{ij}\Big(\sum_{h>i}\sum_{k<j} \pi_{hk}\Big), \tag{2.8}$$

the factor 2 because either observation may be the one in cell (i, j).

2.4.4 Ordinal measure of association: gamma

Definition (Goodman–Kruskal gamma)

Given that a pair is untied on both variables, $\Pi_c/(\Pi_c + \Pi_d)$ is the probability of concordance. *Gamma* is

$$\gamma = \frac{\Pi_c - \Pi_d}{\Pi_c + \Pi_d}, \qquad \text{sample version } \hat{\gamma} = \frac{C - D}{C + D}. \tag{2.9}$$

Proposition 2.18 (Properties of gamma)

(a) $-1 \leq \gamma \leq 1$; $\gamma = 1$ iff $\Pi_d = 0$ and $\gamma = -1$ iff $\Pi_c = 0$ (perfect monotonicity, not linearity).
(b) Reversing the category order of one variable changes the sign of γ; γ is symmetric in X

and Y.
(c) Independence implies $\gamma = 0$; the converse is false.

Proof. [Supplement]

Step (a). $\Pi_c, \Pi_d \geq 0$, so $|\Pi_c - \Pi_d| \leq \Pi_c + \Pi_d$, with the extreme cases as stated.

Step (b). Reversing one ordering turns concordant pairs into discordant ones and vice versa, swapping Π_c and Π_d. Interchanging the roles of X and Y does not change whether a pair is concordant.

Step (c). Let (X_1, Y_1), (X_2, Y_2) be independent draws. Then $\Pi_c = 2P(X_1 > X_2, Y_1 > Y_2)$ and $\Pi_d = 2P(X_1 > X_2, Y_1 < Y_2)$. Under independence of X and Y, (X_1, X_2) is independent of (Y_1, Y_2), so $\Pi_c = 2P(X_1 > X_2)P(Y_1 > Y_2)$ and $\Pi_d = 2P(X_1 > X_2)P(Y_1 < Y_2)$; since Y_1, Y_2 are i.i.d., $P(Y_1 > Y_2) = P(Y_1 < Y_2)$, so $\Pi_c = \Pi_d$ and $\gamma = 0$. For the converse, a U-shaped joint distribution can have $\Pi_c = \Pi_d$ without independence, e.g. the 3×3 table with $\pi_{11} = \pi_{31} = \pi_{22} = 1/3$ has $\Pi_c = 2\pi_{11}\pi_{22} = 2/9 = 2\pi_{22}\pi_{31} = \Pi_d$. ■

2.4.5 Gamma for job satisfaction example

Example 2.19 (Gamma for income and job satisfaction)

Example 2.17: $\hat{\gamma} = (1331 - 849)/(1331 + 849) = 0.221$, a weak tendency for satisfaction to increase with income.